

# ***Take-Home Assignment (due 1159pm ET Wed Apr 29)***

- **This is NOT a group assignment; do not consult another human (or AI app) except me, Zara, or any ERAU-sanctioned tutoring service**
- **There are 3 problems, each of which is worth 33 total points (+ 1 free point)**
- **Show all of your work on each problem to get full credit (just as you've done on exams and HWs)**
- **Write and circle your answers to each problem (so I don't have to hunt for them)**



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**PROBLEM 1:** The year is 2029 and you are now a contractor at Lockheed Martin specializing in structural design and analysis. Your boss has asked you to get creative and think of a unique thin-walled (non-idealized) cantilevered beam scenario that involves bending stress and shear flow analysis. Therefore, you are to:

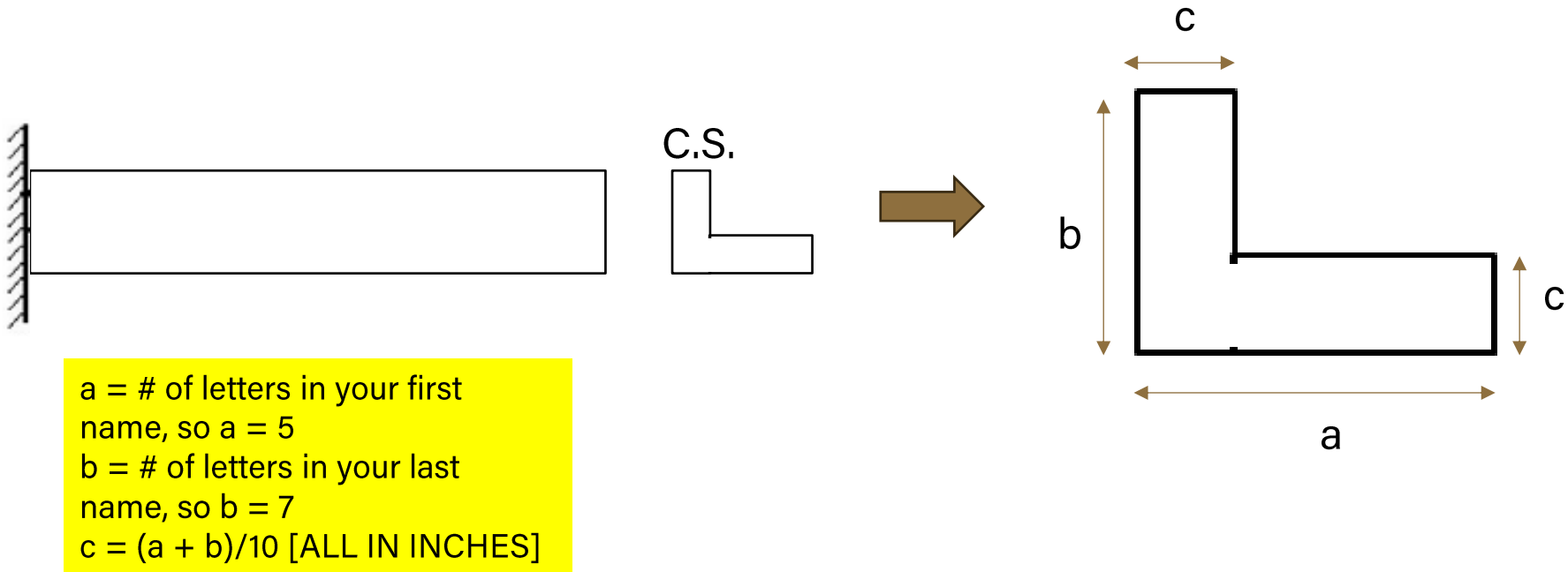
- Choose the length of your beam
- Choose a transverse load (with nonzero  $V_y$  and  $V_z$  components) and the location along the beam where it acts
- Choose your own cross-section (must be different than any of the shapes we've covered in class)
- Choose the material limits of the beam ( $F_u$ ,  $F_y$ ,  $F_{su}$ ,  $F_{sy}$ , in units of psi)

**For the above chosen values, solve the maximum bending stress and the location  $(x, y, z)$  where it occurs, maximum shear flow and the location  $(x, y, z)$  where it occurs, and demonstrate whether the beam is predicted to fail given its material limits.**



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**PROBLEM 2:** Consider the following cantilevered beam and cross-section, with dimensions given as follows:



Suppose we wish to apply a pure moment (i.e. concentrated couple) at the free end of the beam in any direction (i.e. along any axis) that we choose. In particular, we would like to apply the moment such that the neutral axis aligns with the direction of the moment vector. That is, we want the axis about which we apply the moment to also be the neutral axis. Is this possible for the given cross-section? If so, draw the cross-section with the direction of the moment vector clearly indicated (i.e. with a slope, angle, etc) and if not, explain why not.

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**PROBLEM 3:** Work problem 5.5 in the textbook, with the following changes/edits:

- The applied load is a positive vertical ( $V_y$ ) force at the right end of the beam of magnitude  $10*a$
- The radius of semicircle CD is  $a$
- The radius of semicircle DE is  $b$

$a$  = # of letters in your first name, so  $a = 5$   
 $b$  = # of letters in your last name, so  $b = 7$



- 5.5** The beam below consists of webbing material and six stringers. The stringers have already been idealized; i.e., their areas are effective areas. The 10 lb/in line load is applied along the entire 140" span.
- Give the sign & magnitude of  $V_y$ ,  $V_z$ ,  $M_y$ , &  $M_z$  at the wall that you would use in either the general flexure formula or the general shear flow formula.
  - Find the value of  $\sigma_{\text{Largest}}$  at the wall and indicate in which stringer it occurs.
  - Find the location of the shear center, measured from Point E, at the wall & **SHOW YOUR ANSWER ON A DIAGRAM.**
  - Find the shear flows at the wall & **SHOW YOUR ANSWER ON A DIAGRAM.**

